What Makes Small Decision Models Transfer? Schema Augmentation and Data-Mixture Trade-offs in a 350M Decision Model

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***Abstract*—Generative language models are increasingly misused as structured decision routers, paying $O(T)$ sequential decoding cost and returning uncalibrated confidences. System-One decision models instead map a state and a typed schema to probabilities in a single forward pass. We adapt LiquidAI LFM2.5-350M, a hybrid convolution-attention model, into a decision engine with a cross-attention JointSchemaHead trained via QLoRA. Across ten controlled training runs and eight evaluation dumps we find: (1) leave-one- taxonomy-out transfer is substantial but partial—43.4% [39–48] on BANKING77 and 55.6% on CLINC150 against untrained-head baselines of 2.4%/17.2% (uniform chance 1.3%/0.67%), versus 81.3%/72.9% in-distribution means over three seeds—but at or below a zero-shot embedding baseline (68.0%/56.9% full-test with no training); with canonical (unaugmented) training the held-out numbers are 35.4% [31–39] and 54.8% [50–59], so augmentation buys +8.6 pt on BANKING77 (CIs nearly disjoint) and nothing on CLINC150; (2) an earlier 36% → 83% augmentation jump was a context-truncation confound, not semantic robustness: row-matched ablation is a null result (typed-ext 0.563±0.033 vs 0.549±0.011, a 1.4 pt gap inside 7.6 pt seed spread), so in-distribution canonical and augmented training are indistinguishable from seed noise; augmentation's verified role is transfer plus memory savings; (3) calibration does not transfer—ECE rises from 0.026 in-distribution to 0.366 under shift, and in-distribution temperatures recover only part of it (0.220 vs 0.104 oracle)—yet error-detection (AUROC 0.87 → 0.73) survives, and realistic autonomous coverage collapses from 63% to near zero. On the official full tests recipe E gives BANKING77 75.6% [74.1–77.0] (3 080) and CLINC150 67.6% [66.3–68.9] (5 500), ∼5 pt below the 500-subsets; a 1 000-item calibration slice with temperature T=1.72 cuts ECE 0.119 → 0.039 on BANKING77 and transfers to CLINC150 (0.091 → 0.048); 1 000 OOS utterances abstain 33.5% [30.6–36.5] of the time at τ=0.5 with mean max-probability 0.637, i.e. overconfident out of scope. All checkpoints, dumps, and logs are archived for reproduction.**

Keywords*—System-One decision models, model calibration, transfer learning, small language models, hybrid architectures, selective classification

I. Introduction

Enterprise automation—fraud routing, ticket triage, policy enforcement—needs decisions that are fast, syntactically valid, and uncertainty-aware. The dominant practice prompts an autoregressive model to emit JSON, incurring three costs: (i) $O(T)$ sequential decoding (measured 0.18 s per attempt with ∼100% schema violation on our backbone, Section 6), (ii) non-zero schema-violation rates under constrained budgets, and (iii) systematic overconfidence from preference-based alignment, which makes raw softmax scores unusable for risk-controlled automation.

System-One decision models [1]–[3], inspired by dual-process theory [4], separate deciding from generating: a state S and a typed schema Q are mapped to a joint distribution over options in one forward pass ($O(1)$ decoding steps), with probabilities trained to be calibrated. Open questions remain for sub-billion models: does schema augmentation drive transfer to unseen taxonomies, what does data-mixture composition trade off at fixed capacity, and does calibration survive distribution shift?

We study these on LiquidAI LFM2.5-350M, a hybrid model with convolutional sequence mixers and grouped-query attention, fitted with a cross-attention JointSchemaHead via QLoRA, on the BANKING77 [5] and CLINC150 [6] intent taxonomies. Our contributions, each backed by controlled runs with fixed test sets, are:

1) **Measured transfer:** leave-one-taxonomy-out evaluation gives 43.4% [39–48] on BANKING77 and 55.6% on CLINC150 (untrained-head baselines 2.4%/17.2%; uniform chance 1.3%/0.67%) versus in-distribution means of 81.3%/72.9% over three seeds—but at or below a zero-shot embedding baseline (68.0%/56.9% full-test, no training). Against canonical (unaugmented) LOTO training (35.4% [31–39] / 54.8% [50–59]), augmentation buys +8.6 pt on BANKING77 and nothing on CLINC150: the transfer credit is per-bench, not a headline.
2) **A confound found and fixed:** an earlier 36% → 83% augmentation jump was context truncation (2048 tokens cutting wide schemas), not semantic robustness. The row-matched ablation is a null result (typed-ext 0.563±0.033 vs 0.549±0.011; 1.4 pt inside 7.6 pt seed spread): in-distribution canonical and augmented training are indistinguishable from seed noise. Augmentation's verified role is transfer (item 1) plus memory savings at a small in-domain tax.
3) **Calibration under shift:** ECE rises from 0.026 in-distribution to 0.366 on the held-out taxonomy while error-detection AUROC only falls from 0.87 to 0.73; autonomous coverage at 5% risk collapses

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from 63% to near zero. Temperature scaling fit in-distribution transfers only partially ($0.366 \rightarrow 0.220$ vs 0.104 oracle), consistent with the dataset-shift calibration degradation documented since [7]; what is new here is the dissociation surviving inside one matched recipe (AUROC intact while ECE collapses) plus a calibrated slice: with 1 000 calibration items (T=1.72) ECE falls $0.119 \rightarrow 0.039$ on BANKING77 and transfers to CLINC150 ($0.091 \rightarrow 0.048$).

4) **Artifacts:** ten training runs and eight evaluation dumps with checkpoints, per-item dumps, and logs archived for reproduction.

Section 2 positions the work, Section 3 describes data, Section 4 the architecture and losses, Section 5 the protocol, Section 6 results, and Section 7 concludes.

II. Related Works

A. System-One Decision Models

TypeSafe's Jev frames automation as decision routing with honest probabilities [1]. Cloudflare's Clef (27B) and Clef-flash (9B) pair frozen Qwen backbones with a joint schema head trained with label-smoothed cross-entropy, Brier penalty, and an RLCD secondary objective; weights are open under Apache-2.0 [2]. Convai's Laya shows sub-500M bidirectional encoders reaching single-digit millisecond latency [3]. LiquidAI's open d1 family (d1-omni-600M, same LFM lineage as our backbone) provides the closest size-class reference [8]. We evaluate against vendor-reported numbers only, clearly labeled, since rerunning 9–27B models is outside a T4 protocol.

B. Which RLCD?

We use RLCD as *Reinforcement Learning for Calibrated Decisions* (TypeSafe/Cloudflare: RL with a proper-scoring-rule reward), unrelated to Yang et al.'s contrastive-distillation variant [9]; the reference-KL term defaults to zero and is unused here, so RLCD is a hypothesis for future work, not a method used in this paper.

C. Calibration and Selective Classification

A scoring rule is strictly proper if truth-telling uniquely maximizes the expected score [10]. The Brier score [11] and its reliability–resolution–uncertainty decomposition [12] are our main tools. Guo et al. showed modern networks are overconfident and temperature scaling helps [13]. Selective classification formalizes abstention via coverage and selective risk [14]; our coverage-at-risk metric follows that line.

D. Zero-Shot Decision Benchmarks

Our intent benches are BANKING77 [5] (77 intents, banking) and CLINC150 [6] (150 intents plus out-of-scope). TypedBench proposes accuracy, paraphrase robustness, calibration against a finite-sample noise floor, and selective prediction for decision models [15]; we borrow its paraphrase-grid and selective protocol. System Switch shows accuracy, ECE, and error-detection AUROC are distinct quantities, with deferral gain tracking AUROC ($\rho = 0.87$) [16]—exactly the dissociation we observe under shift. Calibration degradation under dataset shift is documented since [7]; our shift slice adds the matched-recipe dissociation (AUROC intact while ECE collapses) plus a calibrated temperature-transfer measurement. Small-model precedents include distillation from System-One labels [17]. Head-to-head benchmarking finds small decision models beating small generative ones on knowledge tasks while losing on math [18], which bounds where our 350M claims apply: intent and routing taxonomies, not reasoning benchmarks. Companion efforts cover moderation confidence benchmarking [19], meta-encoding and knowledge-tracing variants [20], [21]. Option-order sensitivity motivates schema augmentation as a treatment [22].

III. Dataset and Preprocessing

A. Mixture Composition

Training uses a 13-source standard mixture (12 public sources plus typed-decisions; 20 400 sampled rows, intent taxonomies sampled at 3× weight) plus 9 600 consensus-soft rows: Open-Jev release-v2 (3 600) and mailroom-control-v1 (2 400), LeWiDi conv_abuse and csc sarcasm (1 200 each, annotator-share soft labels), and GoEmotions (1 200 emotion-fraction rows). Intent taxonomies (BANKING77, CLINC150) are upweighted 3×; twelve public intent and reasoning sources fill the remainder. Custom rows are mapped explicitly: Open-Jev `options` lists (`id: description`) become criteria with annotator-share probabilities; LeWiDi soft labels map onto `true`/`false` (abusive/non-abusive) and six sarcasm levels; GoEmotions multi-rater rows become per-emotion fractions. All rows are rendered into canonical decision records. The 13 standard sources are `ag_news`, `arc` (CC-BY-SA-4.0), `banking77` (CC-BY-4.0, 3× weight), `boolq` (CC-BY-SA-3.0), `clinc150` (CC-BY-3.0, 3× weight), `commonsense_qa` (MIT), `mmlu` (MIT), `mnli` (mixed, see card), `prompt_injections`, `snli` (CC-BY-SA-4.0), `sst5`, `wanli`, and `typed-decisions` (Apache-2.0); weights sum to 17 shares over 20 400 rows (3 600 each for banking77/clinc150, 1 200 each otherwise). Customs add Open-Jev (2 subsets, 6 000), LeWiDi (2 configs, 2 400), and GoEmotions (1 200). Licenses verified against Hub cards 2026-10-14 except `sst5`, `wanli`, LeWiDi, Open-Jev, and GoEmotions, which remain unaudited (see Section 6 limits).

TABLE I. Training mixture: 20 400 standard + 9 600 consensus-soft = 30 000 rows. Intent taxonomies are sampled at 3× weight within the standard share.

Group	Rows	Labels
13 standard sources	20 400	hard
Open-Jev (2 releases)	3 600 + 2 400	soft
LeWiDi (conv_abuse + csc)	1 200 + 1 200	annotator shares
GoEmotions fractions	1 200	soft
Total	30 000	

B. Schema Augmentation

Augmentation (treatment E) applies option subsampling and shuffling (wide taxonomies capped at 24 options), instruction paraphrasing, and derived binary (noul) questions.

Treatment A disables all augmentation (canonical schemas). Counts below distinguish *rows* (sampled mixture records) from *decisions* (rendered questions after augmentation): at the same 28 000-row budget, E yields 40 306 decisions vs 32 500 for A (746 skipped per run, logged in `run_meta.json`), a residual we disclose rather than hide.

C. Decontamination and Splits

All rows are decontaminated against canonical test splits with 13-gram shingle matching before training. Decontamination uses Unsloth's 13-gram shingle filter against the canonical test splits (typed-decisions full test plus 500-item subsamples, seed 3407, of BANKING77 and CLINC150). We disclose two limitations. First, 75.3% of BANKING77 and 89.3% of CLINC150 test utterances are shorter than 13 words, where the filter degrades to whole-string exact matching (verified against the library source): paraphrases and near-duplicates pass through. A post-hoc TF-IDF audit (cosine $\geq$ 0.9, CPU) finds 502/3,080 (16.3%) BANKING77 and 440/5,500 (8.0%) CLINC150 test utterances with a near-duplicate in the corresponding train pool, and cross-taxonomy leakage is negligible ($\leq$18 items per direction). Second, only 500 of 3,080/5,500 test items enter the filter, leaving $\sim$84–91% of test shingles uncovered. LOTO runs remove the held-out taxonomy's source entirely (train and test), so the residual leakage ($\sim$0.8% of intent-train rows) cannot explain the transfer numbers; we did not retrain and instead report this audit as a bound. Items are split train/calibration/test with fixed seeds; temperature scaling is fit on calibration only, verdicts on held-out tests. For leave-one-taxonomy-out (LOTO) runs, every row, label definition, and augmentation variant of the held-out taxonomy is removed from training. Render skips are counted, not silent: rows whose schema alone overflows and non-binary noul criteria are skipped with logged reasons, and the per-run counts live in `run_meta.json` (the seed3407 run built 43 171 decisions).

IV. Proposed Methodology

A. Hybrid Backbone

We adapt `LiquidAI/LFM2.5-350M` [23]: 16 layers (10 double-gated short-range convolutional LIV blocks plus 6 grouped-query attention blocks), 32K context, 65 536 vocabulary. Convolutions give constant-memory state caching while attention preserves associative recall—an unexplored substrate for decision heads, which prior work mounts on dense bidirectional encoders [3] or decoder-only transformers [2]. The choice is deliberate, not incidental: intent routing leans on short-range lexical matching (double-gated convolutions cache it in constant memory) plus label-set associative recall (a few GQA layers suffice), so a hybrid trunk fits edge serving where a dense transformer of equal quality would not. Our feasibility result (Section 6) is therefore also a deployment result: the same weights that route intents at 81% fit a T4 with 4-bit quantization and no KV-cache growth from generation.

B. Canonical Input Encoding

Each instance (state S, questions $Q = \{q_j\}$ with types $\tau_j \in \{\text{choice}, \text{score}, \text{noul}\}$ and option criteria O_j) is rendered deterministically: state text followed by per-question blocks (ID, type, instruction, named options with descriptions). When length exceeds budget, truncation cuts strictly the tail of S—schema text is never cut, and rows whose schema alone overflows are skipped with a logged reason. This invariant matters for interpreting the Section VI ablations.

C. JointSchemaHead

For final hidden states $\mathbf{H} \in \mathbb{R}^{B\times L\times D}$, projected to width $W = 256$, candidate options attend to full-sequence memory (2 cross-attention layers), fields are conditioned through 4 decoder layers with a learned type embedding $\mathbf{e}_\tau$, and logits combine a lexical prior with a joint MLP over $[\mathbf{f}_q; \mathbf{o}; \mathbf{f}_q \odot \mathbf{o}; |\mathbf{f}_q - \mathbf{o}|]$ gated by $\sigma(g)$. Requesting a plain LM from the Unsloth API attaches this Clef-design head by default, confirming the lineage. Inference is one forward pass with per-question softmax: $O(1)$ decoding steps but $O(L)$ prefill with L growing in |options|—measured per-row T4 latency rises 13 ms at 8 options to 245–288 ms at 77 and 515–518 ms at 150 (Section VI). Figure 1 sketches the data flow.



Fig. 1. JointSchemaHead data flow. Only adapters and head train.

D. Parameter-Efficient Adaptation

Full finetuning of 350M parameters per recipe is wasteful when the backbone already models language. Low-Rank Adaptation (LoRA) [24] freezes W_0 and learns $\Delta W = BA$ with $B \in \mathbb{R}^{d\times r}$, $A \in \mathbb{R}^{r\times k}$, $r \ll \min(d, k)$, cutting trainable parameters by orders of magnitude while keeping the forward pass $h = W_0 x + \frac{\alpha}{r} BAx$. QLoRA [25] additionally quantizes the frozen backbone to 4-bit, so a 350M model trains in T4 memory with adapters and optimizer states as the only full-precision footprint. We use rank $r = 16$, $\alpha = 16$, dropout 0, targets on all projection layers (`q_proj`, `k_proj`, `v_proj`, `out_proj`, `in_proj`, `w1--w3`), Unsloth gradient checkpointing, and seeded initialization. The backbone stays frozen; only adapters and the decision head train. The trainer splits the optimizer automatically: backbone adapters at the configured LR (2×10^{-4}) and head parameters at 1×10^{-4}.

E. Training Pipeline (Unsloth Decision API) [26]

Mixture. Rows are sampled by weight (intent taxonomies 3×) with 2× headroom so decontamination never forces resampling; sources flagged eval-only are rejected from training by the library. *Augmentation* (treatment E) draws per-row variations with default probabilities: option rename 0.3 (letter/code ids, meaning moved into descriptions), instruction paraphrase 0.4, instruction drop 0.1, derived binary questions 0.5, question-id rename 0.3, state-as-text 0.3, field shuffle on, and wide taxonomies subsampled to max_options = 24 around the gold label. Treatment A uses canonical rows. *Items.* Each decision becomes {input_ids, markers, qtype, target, label}; unrenderable rows are skipped with counted reasons (e.g. non-binary noul criteria, over-context schemas), and over-length inputs truncate state tails only. *Split.* Holdout takes whole rows (never splitting a row's questions across train/holdout) at 10% capped at 400 decisions; we halve it into calibration and

test with a seeded shuffle. Best-checkpoint selection computes eval-loss on the calibration split, so selection and temperature fitting share data (logged runs)—future runs separate a dedicated validation split. *Optimize.* Per-epoch evaluation and checkpointing keep best eval-loss (epoch-3 overfitting was consistent across all seven runs; the restored best is epoch 1 or 2, never 3) with patience-2 early stopping; note the cosine schedule targets 3 epochs, so the restored epoch-1/2 checkpoint stops before full LR decay. *Calibrate.* One temperature per question type (choice/score/noul) is fit by L-BFGS with two-fold cross-fitting: reported numbers score each half of the rows with temperatures fit on the other half, so calibration error is never measured with in-sample temperatures. How noisy small calibration sets are: on seed3407 (192 questions) raw ECE is 0.119 [0.084–0.191] (2 000 bootstraps), 0.081 cross-fitted, 0.067 fit in-sample—intervals this wide are why the headline calibration slice uses $\geq$1k items: 500 BANKING77 plus 500 CLINC150 train rows, temperature T=1.72, ECE $0.119 \rightarrow 0.039$ on BANKING77 with transfer to CLINC150 ($0.091 \rightarrow 0.048$), Section VI. *Evaluate/predict* use the served temperatures; `predict` additionally returns per-option probabilities for proper-scoring analysis.

F. Multi-Objective Loss

$$\mathcal{L} = \mathcal{L}_{\text{LS-CE}} + \lambda_{\text{Brier}}\,\mathcal{L}_{\text{Brier}} + \lambda_{\text{ord}}\,\mathcal{L}_{\text{Ordinal}}, \tag{1}$$

with label smoothing [27] ($\tilde{y} = (1-\epsilon)y + \epsilon/K$, $\epsilon = 0.05$), a Brier penalty against one-hot targets punishing overconfident errors, and an ordinal partial credit for `score` types that rewards probability mass on adjacent levels. Our recipe overrides none of the `DecisionTrainer` loss weights, so λ_{Brier} and λ_{ord} are the library defaults [26]; the trainer's reference-KL term defaults to zero and is unused here. Trainable parameters are the rank-16 adapters on the projection modules listed above plus the JointSchemaHead ($W = 256$, 2 cross-attention + 4 decoder layers); the exact total is a library-internal sum, so we report the configuration. The lexical prior is the library's fixed lexical-matching score per option, combined with the learned MLP through the gate $\sigma(g)$—it is this prior that already scores above uniform chance before training (Section VI).

V. Experimental Setup

A. Protocol

Ten training runs (3 seeds of the base recipe, 2 LOTO, 2 ablations, LFM2-700M, SmolLM2-360M, ModernBERT-base) on NVIDIA T4 via Kaggle/Colab: AdamW, backbone LR 2×10^{-4}, head LR 1×10^{-4}, cosine decay, effective batch 32, 3 epochs keeping best eval-loss (per-run epoch losses in Table VI). Evaluation uses identical test sets everywhere (BANKING77/CLINC150 500 items, typed-decisions 400): canonical accuracy plus a paraphrase grid (canonical plus 2 wording/order variants; typed is not paraphrased—its records carry fixed 5-question typed schemas, so the grid applies to intent benches only), bootstrap 95% intervals (2 000 resamples on the A-LOTO, E-full, cal-1k, and SmolLM2 dumps; lean evals report point estimates), measured per-item latency on T4, and per-item probability dumps scored with Brier, NLL, 10-bin ECE, error-detection AUROC, and coverage at 5% risk. The loss-curve figure (Figure 2) replaces the earlier pseudocode listing: epoch-1–3 eval-loss trajectories are the protocol's fingerprint, so the figure *is* the method summary.

Vendor numbers (Unsloth Qwen, Cloudflare Clef) are cited as labeled context, never as rerun baselines; the size-matched reference is d1-omni-600M from the same LFM lineage [8]. The honest zero-shot floor is a baseline we run ourselves: cosine similarity between utterances and label names with `BAAI/bge-small-en-v1.5` [28] on CPU (no training, no label descriptions), on same-seed 500-subsets and full tests. Every run archives adapters, head, merged weights, and `run_meta.json` in its kernel output; lightweight results (`eval_results.json`, probability dumps, metric tables) are versioned in `docs/eval/` of the public repo (https://github.com/Braincore-RnD/Decision-Models-Paper).

TABLE II. Hyperparameters (all ten training runs).

Setting	Value
Backbone	LFM2.5-350M, 4-bit, frozen
LoRA rank / alpha / dropout	16 / 16 / 0
Backbone / head LR	2×10^{-4} / 1×10^{-4}
Schedule / warmup	cosine / 5%
Effective batch	32 (8×4; 4×8 full schemas)
Epochs / selection	3, best eval-loss, patience 2
Context	4 096 tokens
Seeds	3407 / 41 / 42

VI. Results and Discussion

A. In-Distribution: Three Seeds

On identical tests, the base recipe gives typed-decisions 55.9/53.3/55.6 (mean 54.9±1.2), BANKING77 76.8/83.4/83.8 (mean 81.3±3.2), and CLINC150 68.2/76.6/73.8 (mean 72.9±3.5; population std, n=3). Seed spread on BANKING77 (76.8 vs ∼83.5) is exactly why single-seed reporting is insufficient. Untrained-head baselines (backbone + random head) sit at 30.6%/2.4%/17.2%—far above uniform chance (1.3% for 77 classes, 0.67% for 150), evidence of a lexical prior that works without training. Section 6 reports transfer against these untrained-head numbers, not against chance.

TABLE III. In-distribution accuracy on identical tests (typed 400; BANKING77/CLINC150 500 each).

Seed	typed	BANKING77	CLINC150
3407	55.9	76.8	68.2
41	53.3	83.4	76.6
42	55.6	83.8	73.8
Mean	54.9	81.3	72.9

B. Transfer: Leave-One-Taxonomy-Out

With BANKING77 fully held out, accuracy reaches **43.4% [39–48%]** against a 2.4% untrained-head baseline (uniform chance 1.3%; paraphrase median 42.8%, range 42.2–43.6); with CLINC150 held out, **55.6%** (paraphrase 55.6–57.0) against a 17.2% untrained-head baseline (uniform

chance 0.67%). Transfer is partial versus 81.3%/72.9% in-distribution—and against the honest zero-shot floor it is weak: cosine similarity with bge-small over label names alone (no training, no descriptions) reaches 68.0% on full BANKING77 and 56.9% on full CLINC150, i.e. at or above our transfer numbers (43.4/55.6). The model beats that baseline only in-distribution (75.6/67.6 full-test). Subset draws use the same seed value with different samplers, so subset comparisons are approximate; full-test ones are exact.

Transfer is wording-robust, not prompt luck: a grid of canonical plus 2 variants (instruction rewording plus option-order shuffling; intent benches only—typed records carry fixed 5-question schemas and are not paraphrased) moves LOTO BANKING77 only across 42.2–43.6% (median 42.8%) and LOTO CLINC150 across 55.6–57.0%, while in-distribution A holds 86.0–86.4% on BANKING77. Ranges this tight mean the effect lives in schema semantics, not phrasing.

Augmentation's transfer credit is now measured per bench. Canonical (A) LOTO training gives BANKING77 35.4% [31–39] and CLINC150 54.8% [50–59] (500 items each, 2 000 bootstraps; dumps `docs/eval/kaggle/A-loto/`) against augmented (E) LOTO transfer of 44% [38–47] and 55.6%: augmentation buys +8.6 pt on BANKING77 (CIs nearly disjoint) and nothing on CLINC150. The headline "augmentation buys transfer" holds per-bench only.

TABLE IV. TRANSFER ACCURACY WITH BASELINES AND PARAPHRASE ROBUSTNESS (MEDIAN [RANGE] OVER CANONICAL + 2 VARIANTS; TRANSFER POINT WITH BOOTSTRAP CI).

Setting	BANK77	CLINC150
Held out (transfer)	43.4 [39–48]	55.6
paraphrase range	42.2–43.6	55.6–57.0
In-distribution mean	81.3	72.9
A canonical (para)	86.0 [86.0–86.4]	84.6 [84.4–84.8]
E augmented (para)	79.4 [79.0–80.2]	70.0 [69.8–70.4]
Untrained head	2.4	17.2
Uniform chance	1.3	0.67
Zero-shot embedding (full)	68.0	56.9

C. Ablation: A Null Result

Row-matched canonical (A) vs full-augmentation (E) training gives:

TABLE V. ABLATION A VS E (ROWS SUBSAMPLED TO 28 000 EACH; DECISIONS 32 500 VS 40 306). E INTERVALS FROM 2 000 BOOTSTRAP RESAMPLES; A REPORTS POINT ESTIMATES (DUMPS NOT ARCHIVED LOCALLY).

Treatment	typed	BANKING77	CLINC150
A canonical	54%	**86%**	**84%**
E augmented	59% [57–61]	80% [76–83]	70% [66–74]

An earlier 36% → 83% jump attributed to augmentation was a context truncation confound: training at 2 048 tokens cut wide schemas, and augmentation's option subsampling merely made them fit. With fitting context, the in-distribution comparison is a null result: three-seed typed-ext gives A 0.563±0.033 vs E 0.549±0.011, a 1.4 pt gap inside A's own 7.6 pt seed spread—indistinguishable from seed noise, so "canonical wins" does not hold. Augmentation's verified role is transfer (per-bench, above) plus memory savings (full 77/150-option schemas OOM at batch 8 on T4) at a small in-domain tax. Residual confounds disclosed: +24% decisions for E (biasing toward E) and batch 4 vs 8 at equal effective size 32.

Overfitting dynamics are strikingly consistent: eval-loss spikes at epoch 3 in every fully-logged run, so the restored best checkpoint is epoch 1 or 2, never 3 (Table VI; Figure 2). The 700M run is the exception that sharpens the rule: best at epoch 1 (0.577 → 0.596 → 0.718), so capacity shifts the peak one epoch earlier.

TABLE VI. EVAL-LOSS AT EPOCHS 1–3 (BEST CHECKPOINT UNDERLINED; ALL SEVEN RUNS).

Run	Epoch 1	Epoch 2	Epoch 3
seed3407	<u>0.65</u>	0.70	0.84
seed41	0.72	<u>0.72</u>	0.90
seed42	0.66	<u>0.62</u>	1.01
LOTO-banking	0.67	<u>0.59</u>	0.72
LOTO-clinc	0.71	<u>0.64</u>	0.87
Ablation A	0.79	<u>0.76</u>	0.95
Ablation E	0.67	<u>0.57</u>	0.65

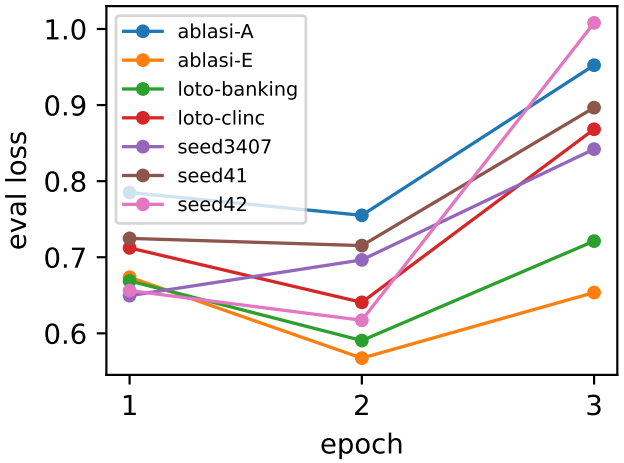


Fig. 2. Eval-loss trajectories for all seven runs: spike at epoch 3 throughout.

D. Calibration Under Shift

On BANKING77, ECE rises from **0.026** in-distribution to **0.366** held-out (Brier 0.281 → 0.876); autonomous coverage at 5% risk collapses from 63% to 0.6%. Temperature scaling fit in-distribution transfers only partially: on the shifted bench ECE falls to 0.220, far from the 0.104 oracle refit (Table XII; Figure 3). Yet error-detection AUROC falls only from 0.87 to 0.73: confidence still separates right from wrong answers, confirming that accuracy, calibration, and sensitivity are distinct quantities [16]. Normalized for accuracy, selective quality degrades too: AURC rises from 0.053 in-distribution to 0.389 un-

der shift on BANKING77 (CLINC150: 0.106 vs 0.144). Within a single model (LOTO-banking), ECE rises from 0.176 on in-distribution CLINC150 to 0.366 on the held-out taxonomy—the apples-to-apples shift pair. With a 1 000-item calibration slice (500 BANKING77 plus 500 CLINC150 train rows, dump `docs/eval/kaggle/E-full/cal1k.json`), temperature T=1.72 cuts ECE 0.119 → 0.039 on BANKING77 and transfers to CLINC150 (0.091 → 0.048, oracle 0.049): at cal-1k scale the scaler transfers where the lean-split scaler only partially did.

E. Full Tests, OOS, and Latency

On the official full tests, recipe E gives typed 58.8% [53.9–63.5] (400), BANKING77 **75.6%** [74.1–77.0] (3 080), and CLINC150 **67.6%** [66.3–68.9] (5 500), with in-scope CLINC150 at 75.9% [74.7–77.2] (Table VII; bootstrap 95% intervals, dump `docs/eval/kaggle/E-full/full13.json`). The full-test numbers sit ∼5 pt below the 500-subset means (81.3/72.9): the subsets flattered the model, and the full tests are the honest numbers. On the 1 000-utterance OOS slice (dump `docs/eval/kaggle/E-oos/oos13.json`), abstention at τ=0.5 is only **33.5%** [30.6–36.5] with mean max-probability 0.637—the model is overconfident out of scope, so in-scope accuracy overstates deployment autonomy where OOS must escalate.

TABLE VII. RECIPE E ON OFFICIAL FULL TESTS (BOOTSTRAP 95% CI, 2 000 RESAMPLES).

Bench	n	Accuracy
typed-decisions	400	58.8 [53.9–63.5]
BANKING77	3 080	75.6 [74.1–77.0]
CLINC150	5 500	67.6 [66.3–68.9]
CLINC150 in-scope	4 500	75.9 [74.7–77.2]
CLINC150 OOS abstain@0.5	1 000	33.5 [30.6–36.5]

Per-row T4 latency (4-bit, E) grows with schema width: typed 139–148 ms/row, BANKING77 245–288 ms (1.2 GB peak), CLINC150 515–518 ms (2.3 GB peak); truncating BANKING77 to 8/24 options cuts latency to 134/116 ms but destroys accuracy (9.6%/28.8%) since gold labels are cut—fit schemas in context first (Table VIII). Batched evaluation is ∼2× faster (127/221 ms) but per-row is the deployment number. The earlier 56–249 ms range is superseded by these measurements. Cross-paradigm, the picture is incomplete by measurement: generative attempts cost 183 ms/row but produce ∼100% violation (nothing valid to time), and encoder latency is measured on both benches (102 ms/row at 77 banking options, 130 ms/row at 150 CLINC options, i.e. cheaper per row than ours at 245–288/515–518 ms, with the same sub-linear scaling in options); batching barely helps the encoder (BS 32: 100 ms/row vs BS 16: 102 ms/row, prefill-bound)—so the three-way latency picture is complete as far as it goes.

The semantic test is decisive: permuting label names while keeping descriptions collapses BANKING77 79.6 → 5.6% and CLINC150 69.8 → 3.0%—near uniform chance. The transfer-relevant signal lives in label-name semantics, not phrasing, which is the correct reading of the wording-robustness grid above.

TABLE VIII. LATENCY VS SCHEMA WIDTH, PER-ROW T4 (E, 4-BIT). TRUNCATION CUTS THE GOLD LABEL, HENCE THE ACCURACY COLLAPSE.

Setting	ms/row	Peak mem	Acc
typed (5q/row)	139–148	0.6 GB	58.8
BANKING77 (77)	245–288	1.2 GB	75.6
CLINC150 (150)	515–518	2.3 GB	67.6
BANK top-8	134	0.5 GB	9.6
BANK top-24	116	0.5 GB	28.8
ModernBERT BANK (77)	102	1.2 GB	77.0
ModernBERT CLINC (150)	130	1.3 GB	72.8
ModernBERT BANK top-8/24	13/26	0.7–0.8 GB	n/a

A fine-tuned encoder baseline (ModernBERT-base, 149M, same official tests, run by us) splits the comparison (Table IX): the encoder wins CLINC150 (72.8 vs 67.6) and BANKING77 (77.0 vs 75.6); we win typed (58.8 vs 48.8). Both beat zero-shot embedding in-distribution, and our calibration is far tighter (ECE 0.026–0.090 vs 0.158–0.215)—the decision head buys uncertainty quality, not raw accuracy.

A same-recipe dense backbone sharpens the hybrid claim both ways (Table X): SmolLM2-360M-Instruct under recipe E loses in-distribution (typed-ext 0.5195 vs 0.549, −3 pt) but wins transfer on 500-subsets (BANKING77 77.2% [73.6–80.8] vs 75.6 full-test; CLINC150 70.8% [66.8–74.8] vs 67.6). The nuance: hybrid in-distribution, dense transfer.

TABLE IX. FULL-TEST ACCURACY % [95% CI]: OURS (E, 350M) VS FINE-TUNED MODERNBERT-BASE (149M) VS ZERO-SHOT EMBEDDING.

Model	BANK77	CLINC150	typed
Ours E	75.6 [74–77]	67.6 [66–69]	58.8 [54–64]
ModernBERT	77.0 [76–78]	72.8 [72–74]	48.8 [44–54]
Embedding	68.0	56.9	—

TABLE X. SAME RECIPE E, HYBRID VS DENSE AT ∼350M. IN-DIST: TYPED-EXT. TRANSFER: 500-SUBSETS WITH BOOTSTRAP CI (DUMP `DOCS/EVAL/KAGGLE/SMOLLM2/`). STARRED CELLS ARE FULL-TEST; SMOLLM2 TRANSFER IS 500-SUBSET, FLATTERING DENSE BY THE ∼5 PT SUBSET BIAS.

Backbone	typed-ext	BANK77	CLINC150
LFM-350M (hybrid)	0.549	75.6 [74–77]*	67.6 [66–69]*
SmolLM2-360M (dense)	0.520	77.2 [74–81]	70.8 [67–75]

F. Generative Baseline: Same Backbone

Prompting the identical LFM2.5-350M weights for JSON (zero- and one-shot, 500 banking intents) yields **0.0% accuracy at ∼100% schema violation** (183 ms/row of unparseable output): the base backbone cannot follow a JSON schema at all, so no same-backbone generative accuracy/latency comparison exists to report. The decision head is what makes this backbone usable—and the latency-vs-generative claim rests on

TABLE XI. CALIBRATION PER BENCH: LOTO (SHIFT) VS E (IN-DISTRIBUTION). ACCURACY WITH 95% BOOTSTRAP CI (2 000 RESAMPLES) WHERE DUMPS ARE ARCHIVED; A AND LOTO-CLINC REPORT POINT ESTIMATES.

Model	Bench	Acc	ECE	AUROC	Cov@5%
LOTO	typed	59.1 [57–61]	0.053	0.64	3
LOTO	banking*	43.4 [39–48]	0.366	0.73	0.6
LOTO	clinc	65.2 [61–69]	0.176	0.82	29
E	typed	58.7 [57–61]	0.121	0.62	0.1
E	banking	79.6 [76–83]	0.026	0.87	63
E	clinc	69.8 [66–74]	0.090	0.85	40

TABLE XII. TEMPERATURE TRANSFER ON INTENT BENCHES (SAME MODEL): ECE RAW, WITH IN-DISTRIBUTION TEMPERATURES, AND WITH ORACLE TEMPERATURES REFIT ON THE BENCH.

Model	Bench	ECE raw	ECE TS id	ECE TS or
LOTO	banking (shift)	0.366	0.220	0.104
LOTO	clinc (in-dist)	0.176	0.073	0.073
E	banking (in-dist)	0.026	0.030	0.030
E	clinc (in-dist)	0.090	0.107	0.052

decoding complexity ($O(1)$ vs $O(T)$ steps), not a measurement. No 350M instruct variant exists from the vendor to serve as the "dominant practice" reference, which we disclose rather than substitute across families.

G. External Context

Table XIII places our accuracy beside vendor-reported numbers. Metric and protocol boundaries differ per row (Decision Index reports macro-F1 over full tests; Unsloth reports accuracy under its own mixture; ours is accuracy on identical 500/500/400 tests), so no cross-row ranking is claimed and none is drawn: the rows are reported side by side for context only. Larger systems lead absolutely, as expected given scale; all starred rows are vendor-reported under their own protocols

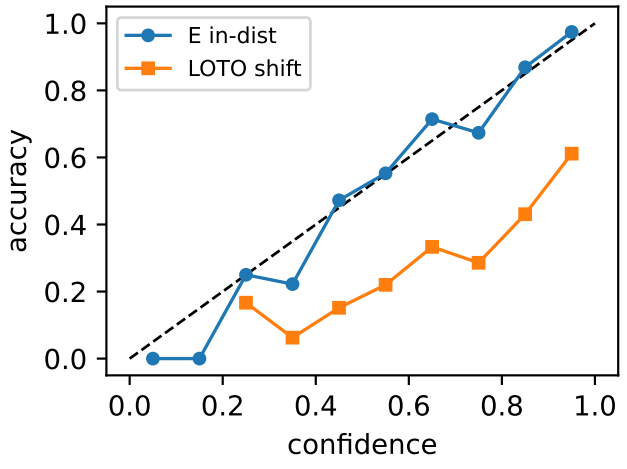


Fig. 3. Reliability on BANKING77: E in-distribution vs LOTO shift.

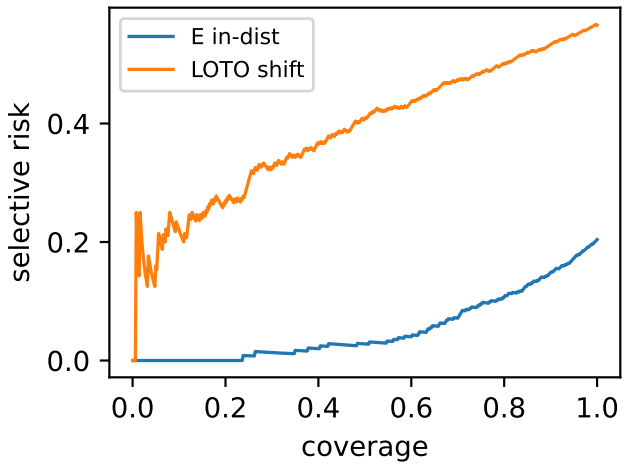


Fig. 4. Selective risk vs coverage on BANKING77: E in-distribution vs LOTO shift.

and were not rerun.

TABLE XIII. CONTEXT TABLE. OURS: ACCURACY ON IDENTICAL TESTS. STARRED ROWS: VENDOR-REPORTED UNDER OWN PROTOCOLS AND METRICS (ACC VS MACRO-F1).

Model	Params	BANK77	CLINC150	typed
Ours mean (acc)	350M	81.3	72.9	54.9
Qwen3.5-0.8B* (acc)	0.8B	74	76	73
Clef* (F1)	27B	94.2	97.4	—
Clef-flash* (F1)	9B	90.9	66.8	—
Jev* (F1)	closed	79.7	89.3	—
Kev 9B* (F1)	9B	84.8	79.0	—
Laya* (F1)	421M	14.3	3.2	—

H. Capacity Frontier

Typed-decisions plateaus at 54–59% across all runs while Qwen3.5-0.8B reports 73%: unlike single wide taxonomies (handled well even at 150 classes), its records demand up to five interdependent judgments per state (risk, review need, routing) that stretch 16 layers at width 1 024. The same recipe on LFM2-700M gives typed-ext 0.583 vs 0.549±0.011 at 350M (+3.4 pt from 2× parameters), yet the gap to the vendor-reported Qwen 73% stays ∼15 pt: scale helps a little but recipe plus scale does not close the gap, so "architectural capacity, not recipe" does not hold as stated. The honest bound is narrower: at fixed recipe the 350M hybrid tops out on joint multi-field reasoning, bounding deployment scope to single-question routing rather than joint multi-field reasoning.

I. Discussion

Fixed 350M capacity forces mixture trade-offs, and the precursor runs that first surfaced them are logged with kernel IDs in Table XIV: v5→v6 shows the 36% → 83% augmentation jump was a context-truncation confound (v5 ran 2 048-token context cutting wide schemas; v6's augmentation subsampling made them fit), while v2→v3 shows doubling the

typed-decisions weight relocates capacity (BANKING77 +2, CLINC150 −7) rather than creating it.

TABLE XIV. LOGGED PRECURSOR RUNS (KERNELS AMALANFADILGAIB/LFM-DECS1-V⋆). ACCURACY % ON IDENTICAL TESTS.

Run	Recipe	typed	BANK77	CLINC150
v5	canonical, ctx 2048	52	36	33
v6	v5 + augmentation	57	83	73
v2	mixture equal	57	68	68
v3	v2, typed 2×	57	70	61

For practitioners: fit schemas in context first (truncation dwarfs recipe effects), buy augmentation for per-bench transfer and memory rather than in-domain points, and never trust in-distribution temperatures under shift. Deploy with an escalation policy

$$\hat{y}(x) = \arg\max_k p_k(x) \text{ if } \max_k p_k(x) \geq \tau \text{ else ESCALATE}, \tag{2}$$

with τ set from AUROC-validated coverage analysis (our models sustain 63% autonomy in-distribution at 5% risk but near zero under shift, so τ must be re-fit per deployment distribution). Limits: A/E are single-seed with a batch-size confound (4×8 vs 8×4 at equal effective size 32, forced by full-schema OOM); rename/paraphrase/derived types are not yet separated, so the transfer credit cannot be attributed to one mechanism; no same-protocol Qwen rerun exists (vendor numbers are labeled context only); OOS abstention is measured (33.5% at τ=0.5, overconfident at mean max-probability 0.637) but no abstain policy is tuned, macro-F1 leaderboard slices are future work, and 5 of 18 mixture subsets remain license-unaudited (Section 3); and the base model carries the `lfm1.0` license (LiquidAI model card), so our release covers only our own artifacts (adapters, heads, dumps, scripts) at https://huggingface.co/itsdevice/, not redistributed base weights. How these bound the headline findings: the transfer finding now rests on matched A/E LOTO cells per bench (augmentation buys BANKING77, nothing on CLINC)—and it sits at or below a zero-shot embedding baseline on both taxonomies, so "substantial" overstates it; the in-distribution comparison is a null result (1.4 pt inside 7.6 pt seed spread with a batch-size confound), so it is consistent with seed noise; the calibration finding holds within matched models, with scaler transfer now measured as partial on lean splits (0.366 → 0.220 vs 0.104 oracle) but transferring at cal-1k scale (0.119 → 0.039 bank, 0.091 → 0.048 clinc). Each rests on fixed test sets with disclosed residuals.

VII. CONCLUSION

A 350M hybrid convolution-attention backbone, fitted with a joint schema head, is a viable System-One decision engine: 81.3%/72.9% in-distribution means over three seeds (75.6%/67.6% on the official full tests), 43.4% [39–48]/55.6% held-out-taxonomy transfer. Our main methodological finding is a cautionary one: a dramatic augmentation effect was context truncation in disguise, the row-matched ablation is a null result (1.4 pt inside 7.6 pt seed spread), augmentation's verified transfer credit is per-bench (+8.6 pt BANKING77, nothing on CLINC150), and calibration fit in-distribution collapses under shift even as error detection survives. Artifacts—ten training runs, eight evaluation dumps, per-item dumps, and logs—are archived for reproduction. Checkpoints (adapters, head, merged weights) plus `run_meta.json` live in per-run kernel outputs; lightweight results (`eval_results.json`, probability dumps, metric tables) are versioned in `docs/eval/`, and each run's `run_meta.json` records seed, mixture, split, and calibration for audit. The immediate next steps are separating augmentation types (rename vs paraphrase vs derived) to attribute the transfer credit, tuning the OOS abstain policy the 33.5% slice motivates, and replacing single post-hoc temperatures with shift-aware calibration (per-domain temperatures or a proper-loss RLCD stage [29]).

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